Regular, subregular and minimal nilpotent orbits.

0. Introduction

Let $\mathfrak{g} \subset \mathfrak{gl}_n(\mathbb{C})$ be a simple Lie algebra and let $G \subset GL_n(\mathbb{C})$ be a simple (connected) Lie group with $Lie(G) = \mathfrak{g}$. We view both G and $\mathfrak{g}$ as irreducible algebraic varieties.

Let $\mathcal{O}_X$ be the nilpotent orbit of $X \in \mathcal{N}$. We will show that:

$$0 \leq \dim \mathcal{O}_X \leq \dim(\mathfrak{g}) - \operatorname{rank}(\mathfrak{g}).$$

We shall see that $\exists!$ orbit $\mathcal{O}_{reg}$ attaining the above upper bound. Also, $\exists!$ second biggest orbit $\mathcal{O}_{subreg}$ of dimension $\dim(\mathfrak{g}) - \operatorname{rank}(\mathfrak{g}) - 2$. Finally, $\exists!$ $\mathcal{O}_{min}$ of minimal dimension among the non-zero orbits.

$\mathcal{O}_{reg}$

|

$\mathcal{O}_{subreg}$

|

⋮ Fuzzy

|

$\mathcal{O}_{min}$

|

$\{0\}$

1. The dimension of a nilpotent orbit.

For $X \in \mathfrak{g}$, write:

$$\chi_{adx}(t) = \sum_{i=0}^{m=\dim(\mathfrak{g})} (-1)^{m-i} a_i(X) t^i,$$

where each $a_i(X)$ is a homogeneous polynomial of degree $m-i$ in $\mathfrak{g}$.

Since $a_0(X) = \det(adx) = 0 \; \forall \; X \in \mathfrak{g}$, $\exists$ a minimal index $1 \leq l < m$ such that

$$a_l \not\equiv 0 \text{ in } \mathfrak{g}.$$

Define $\ell := \operatorname{rank}(\mathfrak{g})$. We use the polynomial q to define the Zariski-open set:

$$\mathfrak{g}' = \{X \in \mathfrak{g} : a_\ell(X) \neq 0\}.$$

Since $\mathfrak{g}$ is irreducible, $\mathfrak{g}'$ is dense in $\mathfrak{g}$.

It is known that $X \in \mathfrak{g}'$ is semisimple and $\mathfrak{g}^X$ is a CSA. Hence,

$$X \in \mathfrak{g}' \Rightarrow \dim \mathfrak{g}^X = \ell = \operatorname{rank}(\mathfrak{g}).$$

We call by regular any element satisfying $\dim \mathfrak{g}^X = \ell$.

<u>Lemma</u> For $X \in \mathfrak{g}$, $\dim(\mathfrak{g}^X) \geq \ell = \operatorname{rank}(\mathfrak{g})$.

<u>Pf.</u> Let M be the set of $Y \in \mathfrak{g}$ s.t. $\dim(\mathfrak{g}^Y)$ is minimal, that is, s.t. ady has max. rank. Then, M is Zariski-open $\Rightarrow M \cap \mathfrak{g}' \neq \emptyset$. Hence, $\dim \mathfrak{g}^X \geq \ell$. □

<u>Prop.</u> Let $X \in \mathcal{N}$. Then, $\mathcal{O}_X$ is an irreducible variety of dimension:

$$0 \leq \dim \mathcal{O}_X = \dim(\mathfrak{g}) - \dim(\mathfrak{g}^X) \leq \dim(\mathfrak{g}) - \operatorname{rank}(\mathfrak{g})$$

<u>Pf.</u> The orbit map $\theta : G_{ad} \to \mathfrak{g}$ has image $\mathcal{O}_X$. Since G is irreducible, then $\mathcal{O}_X$ is also irreducible. Now, the fibres of θ are the cosets $g G_{ad}^X$ of dimension $\dim(G_{ad}^X)$. Hence,

$$\dim \mathcal{O}_X = \underbrace{\dim G_{ad}}_{\dim(\mathfrak{g})} - \underbrace{\dim G_{ad}^X}_{\dim(\mathfrak{g}^X)} \leq \dim(\mathfrak{g}) - \operatorname{rank}(\mathfrak{g}).$$

□

If $X \neq 0$, let $\{X, Y, H\}$ be a std triple containing X. Consider the weight space decomposition of $\mathfrak{g}$ wrt $\mathbb{C}\langle X, Y, H\rangle \simeq \mathfrak{sl}_2$:

$$\mathfrak{g} = \bigoplus_{i \in \mathbb{Z}} \mathfrak{g}_i \quad \text{where} \quad \mathfrak{g}_i = \{Z \in \mathfrak{g} : [H, Z] = iZ\}$$

Prop. $\dim(\mathcal{O}_X) = \dim(\mathfrak{g}) - \dim(\mathfrak{g}_0) - \dim(\mathfrak{g}_1)$.

Pf. Indeed, from the decomposition of $\mathfrak{g}$ into irreducible $\mathfrak{sl}_2$-modules:

$$\dim \mathfrak{g}^X = \#\text{ of highest weight vectors} = \#\text{ of summands} = \dim \mathfrak{g}_0 + \dim \mathfrak{g}_1. \qquad \square$$

2. The regular nilpotent orbit $\mathcal{O}_{reg}$.

We start with the following result.

Lemma. Let $X \in \mathcal{N}$

(i) $\mathcal{O}_X$ is open and dense in $\bar{\mathcal{O}}_X$. In particular, $\dim(\mathcal{O}_X) = \dim(\bar{\mathcal{O}}_X)$.

(ii) $\bar{\mathcal{O}}_X \setminus \mathcal{O}_X$ is a union of orbits of smaller dimension.

Pf.

(i) From algebraic geometry applied to $\theta: G_{ad} \to \mathcal{O}_X$:

$$\exists\ \phi \neq U \subseteq \mathcal{O}_X \text{ s.t. } U \text{ is open in } \overline{\mathcal{O}_X}.$$

Since G_{ad} acts transitively on $\mathcal{O}_X$, then

$$\mathcal{O}_X = \bigcup_{g \in G_{ad}} g \cdot U$$

$\Rightarrow \mathcal{O}_X$ is open in $\bar{\mathcal{O}}_X$. Hence, $\dim(\mathcal{O}_X) = \dim(\bar{\mathcal{O}}_X)$. Moreover, since $\bar{\mathcal{O}}_X$ is irreducible, then $\mathcal{O}_X$ is dense in $\bar{\mathcal{O}}_X$.

(iii) clearly, $\bar{\mathcal{O}}_X$ is G_{ad}-stable $\Rightarrow \bar{\mathcal{O}}_X \setminus \mathcal{O}_X$ is a union of orbits. Since $\bar{\mathcal{O}}_X \setminus \mathcal{O}_X$ is closed and proper in $\bar{\mathcal{O}}_X$, then $\dim(\bar{\mathcal{O}}_X \setminus \mathcal{O}_X) < \dim(\bar{\mathcal{O}}_X) = \dim(\mathcal{O}_X)$ $\square$

Corollary $\mathcal{O} \leq \mathcal{O}'$ if $\bar{\mathcal{O}} \subseteq \bar{\mathcal{O}}'$ defines a partial order in the set of nilpotent orbits.

Pf. Just need to check anti-symmetry: $\bar{\mathcal{O}} = \bar{\mathcal{O}}' \Rightarrow \mathcal{O}$ is open in $\bar{\mathcal{O}}'$ and $\mathcal{O}'$ is dense in $\bar{\mathcal{O}} \Rightarrow \mathcal{O} \cap \mathcal{O}' \neq \phi \Rightarrow \mathcal{O} = \mathcal{O}'$. $\square$.

We now view $\mathcal{N}$ as an affine variety.

Lemma $\mathcal{N} \subseteq \mathfrak{g}$ is an irreducible subvariety.

Pf. Since $\mathfrak{g} \subseteq \mathfrak{gl}_n(\mathbb{C})$, the abstract Jordan decomposition and the usual Jordan decomposition match. Hence,

$$\mathcal{N} = \mathfrak{g} \cap \mathcal{N}(\mathfrak{gl}_n(\mathbb{C})).$$

Therefore, $\mathcal{N}$ is closed in $\mathfrak{gl}_n(\mathbb{C})$ and hence a subvariety. Now, let $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}$ be a Borel subalgebra of $\mathfrak{g}$. We claim: $\mathcal{N} = G_{ad} \cdot \mathfrak{n}$. Indeed, we have $\mathfrak{n} \subseteq \mathcal{N} \Rightarrow G_{ad} \cdot \mathfrak{n} \subseteq \mathcal{N}$. Now, if $X \in \mathcal{N}$, there exists a Borel subalgebra $\mathfrak{b}'$ s.t. $X \in \mathfrak{b}'$. Since $\mathfrak{b}' = x \cdot \mathfrak{b}$ for some $x \in G_{ad}$, then $X \in G_{ad} \cdot \mathfrak{n}$.

Hence, the morphism of varieties $G_{ad} \times \mathfrak{n} \to \mathcal{N}$ is surjective. Since G_{ad} and $\mathfrak{n}$ are irreducible, then $\mathcal{N}$ is also irreducible. □

Rmk $\mathfrak{n}$ is the Lie algebra of the unipotent subgroup $U \subseteq B$. Since U is connected, then U is an irreducible variety. Moreover, $\operatorname{Exp}: U \to \mathfrak{n}$ is an isomorphism of varieties, so $\mathfrak{n}$ is irreducible as well.

We now characterize the largest nilpotent orbit.

Thm $\exists!$ nilpotent orbit in $\mathcal{O}_{reg}$ in $\mathfrak{g}$ of dimension $\dim(\mathfrak{g}) - \operatorname{rank}(\mathfrak{g})$. Moreover, $\mathcal{O}_{reg}$ is open and dense in $\mathcal{N}$.

Pf. Since $\mathcal{N}$ is closed and has finitely many orbits, then

$$\mathcal{N} = \bigcup_{i=1}^{r} \overline{\mathcal{O}_{x_i}} \text{ for some } x_i \in \mathcal{N}.$$

Now, $\mathcal{N}$ being irreducible implies that $\mathcal{N} = \overline{\mathcal{O}_{x_{i_0}}}$ for some i_0.

Hence, $\mathcal{N} \setminus \mathcal{O}_{x_{i0}} = \overline{\mathcal{O}_{x_{i0}}} \setminus \mathcal{O}_{x_{i0}}$ is a union of nilpotent orbits of smaller dimension, ie: $\mathcal{O} \neq \mathcal{O}_{x_{i0}} \Rightarrow \dim(\mathcal{O}) < \dim(\mathcal{O}_{x_{i0}})$. Thus, $\mathcal{O}_{x_{i0}}$ is the unique orbit with the largest dimension possible. Moreover, $\mathcal{O}_{x_{i0}}$ is open and dense in $\overline{\mathcal{O}_{x_{i0}}} = \mathcal{N}$.

We set $\mathcal{O}_{reg} = \mathcal{O}_{x_{i0}}$. We are left to show that

$$\dim \mathcal{O}_{reg} = \dim(\mathfrak{g}) - \operatorname{rank}(\mathfrak{g}).$$

Indeed, let $H \in \mathfrak{h}$ be the unique solution of $CX = \begin{bmatrix} 2 \\ \vdots \\ 2 \end{bmatrix}$, where C is the Cartan matrix of $\mathfrak{g}$. Hence, $\alpha(H) = 2 \ \forall \alpha \in \Delta$. Write:

$$H = \sum_{\alpha \in \Delta} a_\alpha H_\alpha$$

and set $X = \sum_{\alpha \in \Delta} X_\alpha$, $Y = \sum_{\alpha \in \Delta} a_\alpha Y_\alpha$. It's easy to check that $\{X, Y, H\}$ is a standard triple. So, we may consider the weight decomposition:

$$\mathfrak{g} = \bigoplus_{i \in \mathbb{Z}} \mathfrak{g}_i$$

By the construction of H, it's clear: $\mathfrak{g}_1 = \mathfrak{n}_+$ and $\mathfrak{g}_0 = \mathfrak{h}$. Hence,

$$\dim \mathcal{O}_X = \dim(\mathfrak{g}) - \dim(\mathfrak{g}_0) - \dim(\mathfrak{g}_1) = \dim(\mathfrak{g}) - \operatorname{rank}(\mathfrak{g}).$$

Therefore, we must have $\mathcal{O}_X = \mathcal{O}_{reg}$. (bounded achieved) □.

Coroll. $\dim(\mathcal{N}) = \dim(\mathfrak{g}) - \operatorname{rank}(\mathfrak{g})$

Pf. We have $\mathcal{N} = \overline{\mathcal{O}_X} \Rightarrow \dim(\mathcal{N}) = \dim(\overline{\mathcal{O}_X}) = \dim(\mathcal{O}_X)$. □

Ex. Let $\mathfrak{g} = \mathfrak{sl}_4$. Consider $\{X_{[4]}, Y_{[4]}, H_{[4]}\}$ with respect the partition $[4]$. Then, $H_{[4]} = \operatorname{diag}(3, 1, -1, -3)$. Hence, $\alpha(H) = 2 \ \forall \alpha \in \Delta$, which gives that $\mathcal{O}_{X_{[4]}}$ is the regular orbit by uniqueness of triples sharing the same neutral element. ◇

3. The minimal nilpotent orbit $\mathcal{O}_{min}$

We start off with a simple observation:

Prop. The trivial orbit $\{0\}$ is the unique nilpotent orbit of minimal dimension ($=0$).

Pf. We have $\dim \mathcal{O}_X = 0$ iff $\mathfrak{g}^X = \mathfrak{g}$ iff $X \in Z(\mathfrak{g}) = \{0\}$. $\square$

The following lemma will be useful.

Lemma Let $X \in \mathcal{N}$, $0 \neq k \in \mathbb{C}$. Then $kX \in \mathcal{O}_X$

Pf. Assume $X \neq 0$ and let $\{X, Y, H\}$ be its std triple. For $c \in \mathbb{C}$:

$$\operatorname{Exp} cH \cdot X = \sum_{i=0}^{\infty} \frac{(\operatorname{ad} cH)^i}{i!} \cdot X = X + [cH, X] + \frac{1}{2}[cH, [cH, X]] + \cdots$$

$$= X + 2cX + \frac{1}{2}(2c)^2 X + \cdots = e^{2c} X$$

Then, choose c such that $e^{2c} = k \in \mathbb{C}^\times$. $\square$

Corol $\{0\} \subseteq \overline{\mathcal{O}_X}$ $\forall$ $X \in \mathcal{N}$. (because $T_Z \subseteq T_E \Rightarrow \overline{\mathcal{O}_X}^E \subseteq \overline{\mathcal{O}_X}^Z$). $\square$

Thm Let $X_\theta \in \mathfrak{g}_\theta \setminus \{0\}$, where θ is the highest root wrt Φ^+. Then, $\mathcal{O}_{X_\theta} \subseteq \overline{\mathcal{O}_X}$ $\forall X \in \mathcal{N} \setminus \{0\}$. In particular $\mathcal{O}_{X_\theta}$ is the unique nonzero orbit of minimal dimension.

Pf. Consider $\overline{\mathcal{O}_X}$, $X \in \mathcal{N} \setminus \{0\}$. It suffices to show that

$$X_\theta \in \overline{\mathcal{O}_X}.$$

For $X' \in \overline{\mathcal{O}_X}$, write

$$X' = X'_0 + \sum_{\beta \in \Phi} X'_\beta, \quad X'_0 \in \mathfrak{h}.$$

Set $\Phi(X') = \{\beta \in \Phi : X'_\beta \neq 0\}$. We equip $\Phi(X')$ with the lexicographic order. Since $\Phi(X') \neq \emptyset$, let λ be a maximal element in $\Phi(X')$. Then, $\exists\, H_0 \in \mathfrak{h}$:

$$\beta(H_0) < \lambda(H_0) \quad \forall\ \beta \in \Phi(X') \setminus \{\lambda\}.$$

Now, consider the elements:

$$X'_k = e^{-k\lambda(H_0)} \operatorname{Exp}(kH_0) \cdot X' \in \overline{\mathcal{O}}_X, \quad k \in \mathbb{N}.$$

We have:

$$\lim_{k\to\infty} X'_k = \lim_{k\to\infty} \sum_{\beta \in \Phi(X')} e^{k(\beta(H_0) - \lambda(H_0))} X'_\beta = X'_\lambda.$$

Hence, $X'_\lambda \in \overline{\mathcal{O}}_X$ (because $T_Z \subseteq T_E \Rightarrow \overline{\mathcal{O}}_X^E \subseteq \overline{\mathcal{O}}_X^Z$).

Set $\Phi(\overline{\mathcal{O}}_X) = \bigcup_{X' \in \overline{\mathcal{O}}_X} \Phi(X')$ and let λ_0 be a maximal element in $\Phi(\overline{\mathcal{O}}_X)$.

By the above, $X'_{\lambda_0} \in \overline{\mathcal{O}}_X$. Assume $\lambda_0 \neq 0$.

Exercise: If $\lambda_0 \neq 0$, then $\lambda_0 + \alpha \in \Phi$ for some $\alpha \in \Phi^+$.

Since $\lambda_0 + \alpha$ is a root, we have $\mathfrak{g}_{\lambda_0+\alpha} = [\mathfrak{g}_{\lambda_0}, \mathfrak{g}_\alpha]$. Hence, $\exists$ root vector X_α such that $[X_\alpha, X'_{\lambda_0}] \neq 0$.

Set $X'' = \operatorname{Exp} X_\alpha \cdot X'_{\lambda_0} \in \overline{\mathcal{O}}_X$. Then, $\lambda_0 + \alpha \in \Phi(X'') \subseteq \Phi(\overline{\mathcal{O}}_X)$ with $\lambda_0 + \alpha > \lambda_0$, a contradiction.

We have $\mathcal{O}_{X_0} \subseteq \overline{\mathcal{O}}_X \ \forall X \in \mathcal{N} \setminus \{0\}$. Since the irreducible subvarieties of $\overline{\mathcal{O}}_X \setminus \mathcal{O}_X$ have strictly smaller dimension, then $\mathcal{O}_{X_0}$ has the smallest dimension among the non-zero orbits. $\square$

Set $\mathcal{O}_{min} = \mathcal{O}_{X_\theta}$. We finish this section by computing $\dim(\mathcal{O}_{min})$

Prop. The dimension of $\mathcal{O}_{min}$ equals one plus the number of positive roots not orthogonal to θ.

Pf. From the Jacobi identity, we have that $\mathfrak{g}^{X_\theta}$ is $\mathrm{ad}\mathfrak{h}$-invariant. Thus,

$$\mathfrak{g}^{X_\theta} = (\mathfrak{h} \cap \mathfrak{g}^{X_\theta}) \oplus \left(\bigoplus_{\alpha \in \Pi} \mathfrak{g}_\alpha \cap \mathfrak{g}^{X_\theta} \right).$$

Notice: $\ker\theta = \mathfrak{h} \cap \mathfrak{g}^{X_\theta} \Rightarrow \dim(\mathfrak{h}) - \dim(\mathfrak{h} \cap \mathfrak{g}^{X_\theta}) = 1$

We have $\mathfrak{g}_\alpha \cap \mathfrak{g}^{X_\theta} = \mathfrak{g}_\alpha \ \forall\ \alpha > 0$, as $\alpha + \theta$ is not a root.

For $\beta < 0$, we have $(\beta, \theta) \leq 0$ because $\theta - \beta$ is not a root. Now, if $(\beta, \theta) < 0$, then $\beta + \theta$ is a root. Thus,

$$\mathfrak{g}_\beta \cap \mathfrak{g}^{X_\theta} = \mathfrak{g}_\beta \text{ if if } (\beta, \theta) = 0.$$

Therefore, $\dim \mathcal{O}_{X_\theta} = \dim(\mathfrak{g}) - \dim(\mathfrak{g}^{X_\theta})$

$$= \dim(\mathfrak{h}) - \dim(\mathfrak{h} \cap \mathfrak{g}^{X_\theta}) + \sum_{\beta < 0} \left[\dim(\mathfrak{g}_\beta) - \dim(\mathfrak{g}_\beta \cap \mathfrak{g}^{X_\theta}) \right]$$

$$= 1 + \#\text{ negative roots not orthogonal to } \theta$$

$$= 1 + \#\text{ positive roots not orthogonal to } \theta. \qquad \square$$

Fun fact: $\dim \mathcal{O}_{X_\theta} = 2(\check{h} - 1)$ (Weiqiang Wang 99')

Ex. $\mathfrak{g} = \mathfrak{sl}_4$. With respect to the standard Cartan decomposition, we have $X_\theta = E_{1,4}$. It's easy to see that $E_{1,4} \sim E_{2,1}$. Now, since $E_{2,1} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \oplus [0]^{\oplus 2}$, then $\mathcal{O}_{[2,1,1]}$ is the minimal nilpotent orbit.

$\diamond$

3. The subregular nilpotent orbit.

Let κ be the Killing form on $\mathfrak{g}$. Recall that it induces an isomorphism of G-modules: $i_\kappa : (\mathfrak{g}, Ad) \to (\mathfrak{g}^*, CoAd)$ given by:

$$i_\kappa(X) = \kappa(X, \cdot)$$

In particular, we have $i_\kappa(\mathcal{O}_X) = \mathcal{O}_{i_\kappa(X)}$. Since the orbits under the adjoint action are symplectic varieties, it follows that $\dim \mathcal{O}_X$ is always even.

Hence, all nilpotent orbits in $\mathcal{N} \setminus \mathcal{O}_{reg}$ must satisfy:

$$0 \leq \dim \mathcal{O}_X \leq \dim(\mathfrak{g}) - \operatorname{rank}(\mathfrak{g}) - 2.$$

We shall see $\exists!$ nilpotent orbit achieving this new upper bound. We refer to $X \in \mathfrak{g}$ with $\dim \mathfrak{g}^X = \operatorname{rank}(\mathfrak{g}) + 2$ as a subregular element.

A parabolic subalgebra $\mathfrak{p} \subseteq \mathfrak{g}$ is any subalgebra containing a Borel subalgebra. Fixed a choice of simple roots $\Delta \subseteq \Phi$ and Cartan subalgebra $\mathfrak{h}$, we can construct parabolic subalgebras as follows.

For $J \subseteq \Delta$, let $\mathfrak{p}_J$ be the subalgebra generated by $\mathfrak{h}$ and all root spaces $\mathfrak{g}_\alpha$ such that $\alpha \in \Delta$ or $-\alpha \in J$. Clearly, $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n} \subseteq \mathfrak{p}_J$

Ex. (Minimal parabolic subalgebras). Take $J = \{\beta\}$ and write $\mathfrak{p}_\beta := \mathfrak{p}_J$.
We have: $\mathfrak{p}_\beta = \mathfrak{b} \oplus \mathfrak{g}_{-\beta} = (\mathfrak{h} \oplus \mathfrak{g}_\beta \oplus \mathfrak{g}_{-\beta}) \oplus \left(\sum_{\substack{\alpha > 0 \\ \alpha \neq \beta}} \mathfrak{g}_\alpha \right)$.

Let $\mathfrak{p} = \mathfrak{l} \oplus \mathfrak{u}$ be the Levi factorization of the parabolic subalgebra $\mathfrak{p}$, where $\mathfrak{l}$ is reductive and $\mathfrak{u}$ is the nilradical. Since $\mathfrak{u}$ is the Lie algebra of a unipotent radical, then $\mathfrak{u}$ is an irreducible variety. Moreover, we have $\mathfrak{u} \subseteq \mathcal{N} = \bigcup_{i=1}^{r} \mathcal{O}_{X_i}$.

Hence, we get the decomposition: $\mathfrak{u} = (\mathfrak{u} \cap \mathcal{O}_{X_1}) \cup \dots \cup (\mathfrak{u} \cap \mathcal{O}_{X_r})$. Since $\mathfrak{u}$ is irreducible, at least one $\mathfrak{u} \cap \mathcal{O}_{X_{i_0}}$ is dense in $\mathfrak{u}$.

Thus, we have $u = u \cap \overline{\mathcal{O}_{x_{i_0}}}$. Since $\mathcal{O}_{x_{i_0}}$ is open in $\overline{\mathcal{O}_{x_{i_0}}}$, then $u \cap \mathcal{O}_{x_{i_0}}$ is open and dense in $u \cap \overline{\mathcal{O}_{x_{i_0}}} = u$. Moreover, we have $u \cap \mathcal{O}_{x_j} \subseteq u \setminus u \cap \mathcal{O}_{x_{i_0}}$ for all $j \neq i_0$, so any $u \cap \mathcal{O}_{x_j}$ is not dense and of smaller dimension.

Set $\mathcal{O}_p = \mathcal{O}_{x_{i_0}}$. It is the unique orbit intersecting u in an open and dense set. We refer to $\mathcal{O}_p$ as the Richardson orbit of p; the elements in $u \cap \mathcal{O}_p$ are called the Richardson elements.

We want to compute $\dim(\mathcal{O}_p)$. We need the following Lemma. The proof is quite long, see Carter's book.

Lemma $u \cap \mathcal{O}_p$ is a single P-orbit, where $P \subseteq G$ is a parabolic subgroup with $p = \mathrm{Lie}(P)$. □

Corollary. Let $X \in \mathcal{O}_p \cap u$. Then, $[X, p] = u$.

Pf. We have:

$$\overline{P \cdot X} = u \cap \overline{\mathcal{O}_p} = u.$$

Hence, $\dim u = \dim \overline{P \cdot X} = \dim \overline{[X, p]} = \dim [X, p]$

Since we have $[X, p] \subseteq u$, equality follows □.

Lemma. We have $\dim \mathcal{O}_p = 2 \dim u$.

Pf. Consider the opposite parabolic subalgebra $\bar{p} = l \oplus \bar{u}$, where $\bar{u}$ is generated by $g_{-\alpha}$ for every $\alpha \in \Phi^+$ s.t. $g_\alpha \subseteq u$. Then,

$$g = p \oplus \bar{u}$$

Let $X \in \mathcal{O}_p \cap u$. We show that $\mathrm{ad}X|_{\bar{u}}$ is injective.

Indeed, if we have $ad_X \cdot Y = 0$ for some $Y \in \bar{u}$, then

$$0 = \kappa(ad_X \cdot Y, p) = \kappa(Y, [X, p]) = \kappa(Y, u).$$

Since u and $\bar{u}$ are paired non-degenerately, then $Y = 0$. Hence, $\dim [X, \bar{u}] = \dim \bar{u} = \dim u$. Thus,

$$\dim [X, g] = \dim [X, p] + \dim [X, \bar{u}] = 2 \dim u.$$

Since $\dim [X, g] = \dim g - \dim g^X = \dim \mathcal{O}_X$, the result follows. □

Thm There exists a unique nilpotent orbit $\mathcal{O}_{sreg}$ that is dense in $\mathcal{N} \setminus \mathcal{O}_{reg}$ of dimension $\dim(g) - \operatorname{rank}(g) - 2$.

Pf. For $\beta \in \Delta$, consider $\mathcal{O}_\beta := \mathcal{O}_{p_\beta}$. For the nilradical $u_\beta \subseteq p_\beta$, we have: $u_\beta = \sum_{\alpha \in \Phi^+ \setminus \langle \beta \rangle} g_\alpha$. Hence, $\dim u_\beta = \frac{|\Phi|}{2} - 1$. Therefore,

$$\dim \mathcal{O}_\beta = 2 \dim(u_\beta) = |\Phi| - 2 = \dim(g) - \operatorname{rank}(g) - 2.$$

This shows the existance of nilpotent subregular elements. Now, let $X \in \mathcal{N}$ be nilpotent subregular and let $\{X, Y, H\}$ be its standard triple. By the proof of the regular orbit theorem, we must have

$$\beta(H) \neq 2 \quad \text{for some } \beta \in \Delta.$$

Now we consider the parabolic subalgebra $p_\beta = l_\beta \oplus u_\beta$. Then, we have $X \in u_\beta$. Since $u_\beta \subseteq \bar{\mathcal{O}}_\beta$ by density of $u_\beta \cap \mathcal{O}_\beta$ in u_β, then $\mathcal{O}_X = \mathcal{O}_\beta$. This shows that any nilpotent subregular orbit is induced from a minimal parabolic subalgebra.

In order to show uniqueness, we are left to show:

$$\mathcal{O}_\beta = \mathcal{O}_\alpha \quad \forall \alpha, \beta \in \Delta.$$

See McGovern's book for a proof. We finish by proving that $\mathcal{O}_{sreg}$ is open and dense in $\mathcal{N} \setminus \mathcal{O}_{reg}$. Indeed, any $X \in \mathcal{N} \setminus \mathcal{O}_{reg}$ must must satisfy $X \in u_\beta \subseteq \overline{\mathcal{O}}_{sreg}$ for some $\beta \in \Delta$. Hence, we get $\mathcal{N} \setminus \mathcal{O}_{reg} \subseteq \overline{\mathcal{O}}_{sreg}$. Conversely, any $X \in \overline{\mathcal{O}}_{sreg}$ must have

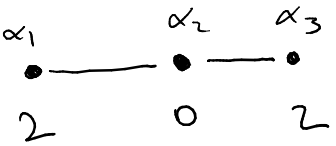
$$\dim \mathcal{O}_X \leq \dim \overline{\mathcal{O}}_{sreg} < \dim \mathcal{O}_{reg},$$

whence $\overline{\mathcal{O}}_{sreg} \subseteq \mathcal{N} \setminus \mathcal{O}_{reg}$. Thus, $\mathcal{O}_{sreg}$ is dense in $\mathcal{N} \setminus \mathcal{O}_{reg}$. □

Ex. $\mathfrak{g} = \mathfrak{sl}_4$. We claim that $\mathcal{O}_{X_{[3,1]}} = \mathcal{O}_{sreg}$. We just need to show $\dim \mathfrak{g}^{X_{[3,1]}} = 3+2 = 5$ by computing its weighted Dynkin diagram:

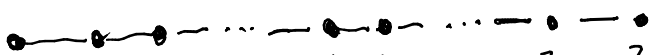
$$h_{[3,1]} = \operatorname{diag}(2,0,-2,0) \sim \operatorname{diag}(2,0,0,-2),$$

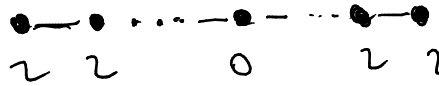
Therefore, $X_{[3,1]}$ has diagram:



We show $\dim \mathfrak{g}_0 + \dim \mathfrak{g}_1 = 5$.

Clearly, $\mathfrak{g}_1 = \mathfrak{sl}_2$ and $\mathfrak{h} \subseteq \mathfrak{g}_0 \Rightarrow \dim \mathfrak{g}_0 \geq 3$. Now, from the diagram, we see that $\mathfrak{g}_{\alpha_2} \subseteq \mathfrak{g}_0$. Since $\mathfrak{g}_{-\alpha_2} \subseteq \mathfrak{g}_0$ as well, then $\dim \mathfrak{g}_0 = 3 + 2 \cdot 1 = 5$.

In general, $X_{[n,1]}$ has diagram 

for n odd and 

for n even. Hence,

$\dim \mathfrak{g}_0 + \dim \mathfrak{g}_1 = n+2$. ◇

Appendix.

1. Let $\mathfrak{g} = \mathfrak{sl}_n$. We show an alternative proof of $\mathcal{O}_\lambda \subseteq \overline{\mathcal{O}_{reg}} = \overline{\mathcal{O}_{[n]}}$, $\lambda \vdash n$.

For $1 \le k \le n-1$, let $Z_k = X_{[n]} - E_{k,k+1}$. We show that $Z_k \in \overline{\mathcal{O}_{[n]}}$. Let $0 \neq c \in \mathbb{C}$ and set $b = c^{-k/n}$, $a = bc$. Consider the following matrix $T_k = \mathrm{diag}(\underbrace{a, \dots, a}_{k \text{ times}}, b, \dots, b)$.

Claim 1 $T \in SL_n$: $\det T = a^k b^{n-k} = b^k c^k b^{n-k} = c^k \cdot b^n = 1$

Claim 2 $T_k \cdot X_{[n]} \cdot T_k^{-1} = Z_k + c \cdot E_{k,k+1}$.

Indeed, the entry $(i, i+1)$ of $T \cdot X_{[n]} T^{-1}$ is

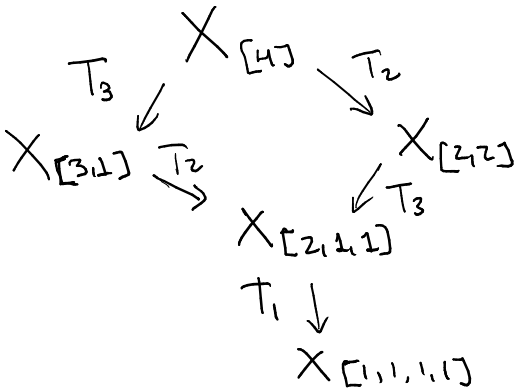
$$\sum_{j,l} T_{ij} X_{jl} T^{-1}_{l,i+1} = \sum_{l} T_{ii} X_{il} T^{-1}_{l,i+1}$$

$$= T_{ii} X_{i,i+1} T^{-1}_{i+1,i+1} = \begin{cases} a \cdot a^{-1} = 1 & \text{if } i < k \\ a \cdot b^{-1} = c, & \text{if } i = k \\ b \cdot b^{-1} = 1, & \text{if } i > k \end{cases}.$$

Therefore, $X_k \in \overline{\mathcal{O}_{[n]}}$ by letting $c \to 0$.

Given $\lambda \vdash n$, it's not difficult to see that we can construct X_λ by a finite sequence of conjugations by the elements T_k.

Ex. $\mathfrak{g} = \mathfrak{sl}_4$.



2. Let $\mathfrak{g} = \mathfrak{sl}_n$. We want to better explore the partial order on orbits.

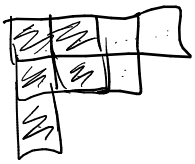
Thm. We have $\mathcal{O}_\lambda \leq \mathcal{O}_\mu$ if/if $\lambda \trianglelefteq \mu$ in the dominance order.

We need a few lemmas.

Lemma 1. $\operatorname{rank}\left(X_\lambda^k\right) = \#$ boxes in λ removed the k first columns.

Pf. **Claim 1:** $\operatorname{rank}\left(X_\lambda^k\right) = \sum_{\{j:\lambda_j \geq k\}} \lambda_j - k$. This is clear for a single Jordan block, so we sum across all blocks.

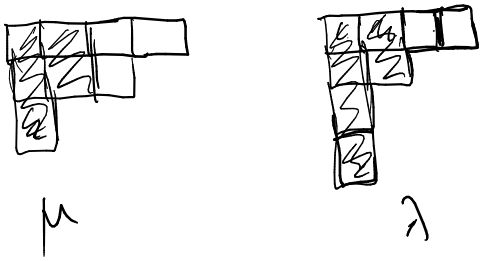
Now, notice that $\sum_{\{j:\lambda_j \geq k\}} \lambda_j - k$ is exactly the number of boxes remaining after removing the first k columns.

Ex. $\mathfrak{g} = \mathfrak{sl}_8$  $\rightsquigarrow \operatorname{rank}\left(X^2_{[4,3,1]}\right) = 2$.

Lemma 2. $\operatorname{rank}\left(X_\mu^k\right) \geq \operatorname{rank}\left(X_\lambda^k\right)$ $\forall k$ if/if $\mu \trianglerighteq \lambda$.

Pf.

Indeed, after removing the first k columns, the remaining boxes of μ is at least as much as in λ if/if $\mu \trianglerighteq \lambda$.



μ λ

Now, let $\mathcal{O}_\lambda \subseteq \overline{\mathcal{O}_\mu}$. Set $m_k = \operatorname{rank}(X_\mu^k)$. Then, the set $\{z : \operatorname{rank}(z^k) \leq m_k\}$ is Zariski closed, because every $l \times l$ minor of z^k with $l > m_k$ has to vanish. Hence,

$$M = \bigcap_{k \geq 0} \{z : \operatorname{rank}(z^k) \leq m_k\}$$

is also Zariski-closed. Now, rank is invariant under conjugation, so M is G-invariant. Moreover, $X_\mu \in M$ by construction. Hence, $\mathcal{O}_\mu \subseteq M \Rightarrow \overline{\mathcal{O}_\mu} \subseteq M$. Since $X_\lambda \in \overline{\mathcal{O}_\mu}$ by assumption, then

$$\operatorname{rank}(X_\lambda^k) \leq \operatorname{rank}(X_\mu^k) \quad \forall k$$

$\Rightarrow \mu \trianglerighteq \lambda$ by Lemma 2.

The converse holds as well. The proof goes by showing that $X_\lambda \in \bigoplus_{i \geq 2} g_i$, where $\bigoplus g_0$ is the weight decomposition wrt the std triple $\{X_\mu, H_\mu, Y_\mu\}$. By showing that $\mathcal{O}_\mu$ is dense in $\bigoplus_{i \geq 2} g_i$, the result follows.